CCW® Armored Power, 15 kV 133%, Shielded, 3/C

UL Type MC-HL or MV-105, EPR, 105°C, Cable Tray Use, Sunlight-Resistant
Direct Burial, ABS CWCMC



Product Construction:

Conductor:
- Bare annealed copper per ASTM B3
- Compact stranding per ASTM B496

Extruded Strand Shield (ESS):
- Extruded thermoset semi-conductor stress control layer over the conductor per ICEA S-93-639 and UL 1072

Insulation:
- 220 mils Ethylene Propylene Rubber (EPR) insulation per ICEA S-93-639 and UL 1072

Extruded Insulation Shield (EIS):
- Thermoset semi-conducting polymeric layer, free stripping from the insulation per ICEA S-93-639 and UL 1072

Shield:
- 5 mil annealed bare copper tape with 25% overlap

Phase Identification:
- Color-coded polymeric identification tape laid under the shield - black, red and blue

Grounding Conductor:
- Class B stranded bare annealed copper grounding conductor
- Sized in accordance with UL 1072 and NEC Article 250

Cable Assembly:
- Insulated and grounding conductors are cabled together with non-hygroscopic fillers when required
- Binder tape is applied over the cabled core

CCW Armor:
- Impervious, continuously welded and corrugated aluminum alloy sheath per UL 1072 and UL 1569
- CCW armor conductivity meets the grounding requirements of the NEC

Jacket:
- Flame-retardant, moisture- and sunlight-resistant Polyvinyl Chloride (PVC), red
- Low temperature performance meets ASTM D746 brittleness temperature at or below -40°C

Applications:

- CCW armored Medium-Voltage power cables offer an economical, rugged and reliable alternative to labor-intensive cable in conduit wiring methods
- For use on feeders and branch circuits in industrial power distribution systems per NEC Articles 328 and 330
- For use in Class I, II and III, Divisions 1 and 2; and Class I, Zones 1 and 2 hazardous locations per NEC Articles 501, 502, 503 and 505
- Installed on metal racks, troughs, in raceways, in cable trays or secured to supports spaced not more than six feet apart
- Installed in both exposed and concealed work, wet or dry locations, directly buried or embedded in concrete
- Recognized for use on fixed or floating offshore petroleum facilities as recommended by the American Petroleum Institute

Features:

- CCW armor provides an impervious barrier to moisture, gas and liquids and meets the grounding requirements of UL 1072 and the NEC
- Triple Extrusion: The strand shield, EPR insulation and insulation shield are all extruded in one operation

Features: (cont'd)

- General Cable's EPR insulation system has outstanding corona resistance and high dielectric strength, and it provides electrical stability under stress
- Cable meets cold impact at -40°C
- 105°C continuous operating temperature, wet or dry
- 140°C emergency rating
- 250°C short circuit rating

Specifications:

Design Adherence:
- ICEA S-93-639/WC-74, 5-46 kV Shielded Power Cable
- AEIC CS8 Specification for Shielded Power Cable, 5-46 kV
- UL 1072 Medium-Voltage Power Cables
- UL 1569 Metal Clad Cables
- UL 2225 Cables and Cable Fittings for Use in Hazardous Locations
- UL 1309 Marine Shipboard Cable

Flame Tests:
- ICEA T-29-520 (210,000 BTU/hr)
- IEEE 383 (70,000 BTU/hr)
- CSA FT4
- IEEE 1202 (70,000 BTU/hr)
- UL 1072
- IEC 60332-3 Category A

Compliances:
- UL Type MV-105, UL File # E90501
- UL Type MC-HL, SUN RES, CT USE, DIR BUR, -40°C, FT4, UL File # E90496
- UL Listed Marine Shipboard, UL File # E85994
- American Bureau of Shipping (ABS) Listed for CWCMC

General Cable
Phone: 888-593-3355
www.generalcable.com

ABS
UL LISTED

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PART NUMBER	COND. SIZE	NO. OF COND.	INSULATION THICKNESS		NOMINAL O.D. OVER INSULATION		BARE GROUND	NOMINAL CORE O.D.		NOMINAL ARMOR O.D.		JACKET THICKNESS		NOMINAL OVERALL O.D.		APPROXIMATE NET WEIGHT		AMPACITY	
	AWG (kcmil)		mils	mm	INCHES	mm	AWG	INCHES	mm	INCHES	mm	mils	mm	INCHES	mm	LBS/ 1000 FT	kg/ 1000 m	IN AIR[1]	DIRECT BURIAL[2]
3/C WITH GROUND MC-HL OR MV-105, 220 MILS EPR, 15 kV 133% INSULATION LEVEL																			
9835.00203106	2 (7/W) (33.6 mm²)	3	220	5.6	0.76	19.3	6	1.82	46.2	2.15	54.6	60	1.52	2.28	57.9	2,473	3,680	185	200
9835.00103104	1 (19/W) (42.4 mm²)	3	220	5.6	0.79	20.1	4	1.88	47.8	2.23	56.6	60	1.52	2.36	59.9	2,811	4,183	210	225
9835.11003104	1/0 (19/W) (53.5 mm²)	3	220	5.6	0.83	21.1	4	1.96	49.8	2.32	58.9	75	1.91	2.48	63.0	3,190	4,747	240	255
9835.21003104	2/0 (19/W) (67.4 mm²)	3	220	5.6	0.87	22.1	4	2.06	52.3	2.40	61.0	75	1.91	2.56	65.0	3,630	5,402	275	290
9835.41003103	4/0 (19/W) (107 mm²)	3	220	5.6	0.97	24.6	3	2.26	57.4	2.62	66.5	75	1.91	2.79	70.9	4,435	6,600	360	345
9835.25003102	250 (37/W) (127 mm²)	3	220	5.6	1.03	26.2	2	2.36	59.9	2.75	69.9	75	1.91	2.92	74.2	5,086	7,569	400	410
9835.35003102	350 (37/W) (177 mm²)	3	220	5.6	1.12	28.4	2	2.61	66.3	3.03	77.0	85	2.16	3.21	81.5	6,445	9,591	490	495
9835.50003101	500 (37/W) (253 mm²)	3	220	5.6	1.24	31.5	1	2.86	72.6	3.32	84.3	85	2.16	3.50	88.9	8,376	12,465	600	590
9835.750031110	750 (61/W) (380 mm²)	3	220	5.6	1.41	35.8	1/0	3.25	82.6	3.80	96.5	85	2.16	3.98	101.1	11,431	17,011	745	720

Dimensions and weights are nominal; subject to industry tolerances.
[1] Ampacities in air are per NEC Table 310.60(C)(71) for an insulated three-conductor copper cable isolated in air with 105°C rated conductors at a 40°C ambient temperature.
[2] Direct burial ampacities are per NEC Table 310.60(C)(83) for three insulated copper conductors cabled within an overall covering, directly buried in earth with 105°C rated conductors at 20°C ambient earth temperature.